

1.6 Writing Equivalent Expressions Using Rational Exponents

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.		 Learning Targets	- I can apply the Laws of Exponents to rational exponents. - I can convert between expressions involving rational exponents and expressions involving radicals.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.		N/A	
 Essential Questions	- How can I use the relationship between rational exponents and radical expressions to convert between the two forms? - How can the Laws of Exponents be used to convert between rational exponent form and radical expression form?			
 Lesson Narrative	In this lesson, students will continue using the Laws of Exponents to convert between rational exponent form and radical expression form. Students will receive a greater amount of exploration and practice in this lesson, giving them the opportunity to reinforce what they have been learning in the last several lessons and to improve their fluency.			
 Component Summary	1.6.1 Warm-Up (~10 min.)	Identify equivalent expressions matching activity (<i>SE p. 33</i>)	1.6.4 Activity (5-10 min.)	Card sort activity with expressions written in rational exponent form and radical expression form into equivalent sets (<i>SE p. 34</i>)
	1.6.2 Guided Practice (5 min.)	Writing equivalent expressions practice (<i>SE p. 33</i>)	1.6.5 Try It (5-10 min.)	Practice writing equivalent expressions in rational exponent and radical expression form with a partner (<i>SE p. 35</i>)
	1.6.3 Exploration (5-8 min.)	Match expressions in rational exponent form to radical expression form (<i>SE p. 34</i>)	1.6.6 Your Turn! (5 min.)	Practice rewriting radical expressions into rational exponent form independently (<i>SE p. 35</i>)
	1.6.7 Wrap-Up (~5 min.)	Cool down that covers equivalent expressions in both radical and rational exponent form (<i>SE p. 36</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		- Cards for Card Sort Activity - Calculators	Work out each of the activities on your own prior to planning for the lesson. This will help you get an understanding of the activities to best pace your lesson accordingly.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not make the connection between the two forms. - Students may put the numerator of the fraction in the exponent as the index of the radical form. - Students may make calculation errors when determining the value of the expression.	- This lesson has a lot of collaborative activities. Consider your student learning styles and specific classroom needs when pacing out the activities and discussions. - Consider your students to determine if having a calculator for 1.6.1 Warm-Up will help students determine equivalent expressions. The focus should be on relating the equivalent expressions, and for some students the different bases may be a struggle for them. - Calculators are used throughout the lesson to help support students learning equivalent rational exponents and radical expressions. - If time is an issue, then consider assigning 1.6.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During the collaborative activities of this lesson, students can engage in a Think-Pair-Share protocol. Consider leveraging the Think-Pair-Share protocol to allow students to work with a partner after they've done it on their own and discuss their methods and findings.	MA.K12.MTR.4.1: Discussion and Reflection Students will work collaboratively for most of this lesson. The collaboration will require students to engage in discussions that require them to reflect on their own thinking and the thinking of others.
 Differentiate Instruction	Student performance on 1.6.5 Try It will provide data on where students are struggling to be able to provide them differentiated and targeted support during 1.6.6 Your Turn!. For 1.6.4 Activity, consider creating physical cards for the activity to allow kinesthetic learners to have a more hands-on experience with this activity. Students will complete a Ticket Out the Door that assesses students' understanding of equivalent expressions in radical expression in rational exponent form.	
Lesson Notes:		

1.6.1 Warm-Up: Identifying Equivalent Expressions

Component Overview: Students will apply prior knowledge to identify equivalent expressions. This activity is designed to have students complete question 1 on their own and then work with a partner to complete questions 2 and 3.



1.6.1 Warm-Up: Identifying Equivalent Expressions

Three lists of numbers are shown.

List A	List B	List C
$8 \cdot 8 \cdot 8$	2^9	24
$4 \cdot 16 \cdot 2$	$6 \cdot 4$	$\frac{1}{32}$
$2 \cdot 16$	$\left(\frac{1}{2}\right)^5$	512
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$2^2 \cdot 2^3$	128
$4 + 4 + 4 + 4 + 4 + 4$	$2 \cdot 4^3$	32



Students may need additional support with the directions of this activity. Consider completing one to show students what to do or provide them explicit instructions on what to do on their own and how many matches each list should have and not have.

1. Choose an expression from List A and match it with an equivalent expression from List B and List C.

$8 \cdot 8 \cdot 8$	2^9	512
$4 \cdot 16 \cdot 2$	$2 \cdot 4^3$	128
$2 \cdot 16$	$2^2 \cdot 2^3$	32
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$\left(\frac{1}{2}\right)^5$	$\frac{1}{32}$
$4 + 4 + 4 + 4 + 4 + 4$	$6 \cdot 4$	24

2. Explain to your partner how you know the expressions you matched are equivalent.

Answers will vary.

3. Listen to your partner explain the match they found. If you disagree, discuss your thinking and work to reach an agreement.

Answers will vary.

1.6.2 Guided Practice: Writing Equivalent Expressions

Component Overview: Students will practice writing equivalent expressions through a structured format using the Laws of Exponents.



1.6.2 Guided Practice: Writing Equivalent Expressions

- Use the Power of a Power Law of Exponents to write an equivalent exponential expression for the following two expressions.

a. Expression 1: $\left(a^{\frac{1}{c}}\right)^b = a^{\left(\frac{1}{c} \cdot b\right)} = a^{\frac{b}{c}}$

b. Expression 2: $(a^b)^{\frac{1}{c}} = a^{(b \cdot \frac{1}{c})} = a^{\frac{b}{c}}$

- c. Explain why the expressions $\left(a^{\frac{1}{c}}\right)^b$ and $(a^b)^{\frac{1}{c}}$ are equivalent.

Answers will vary. They are equivalent because when you multiply the exponents together, they result in the same final value. They are equivalent because the numerator is 1, when they are multiplied, the values are the same.

- Two students evaluated an expression using two different methods.

Jerry	Antonio
$\left((81)^{\frac{1}{2}}\right)^3$	$\left((81)^{\frac{1}{2}}\right)^3$
$(\sqrt{81})^3$	$81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}}$
$(9)^3$	$\sqrt{81} \cdot \sqrt{81} \cdot \sqrt{81}$
729	$9 \cdot 9 \cdot 9$
	729

Explain the difference between the two methods.

Jerry converted from exponent form to radical form, simplified, and then evaluated. Antonio rewrote the power of a power rule in expanded form, then converted from exponent to radical form, and then evaluated.



After question 1 part c, allow students to share out their thinking around why the expressions are equivalent. Consider choosing different input values to show these expressions being evaluated with a numeric value to show their equivalency.



- How do you think it's possible to be able to evaluate the same expression using two different methods and still get the same answer?
- How does the relationship between rational exponents and radical form allow that to happen for Jerry and Antonio?

1.6.3 Exploration: Matching Expressions

Component Overview: Students will work with a partner to match expressions in rational exponent form to radical expression form. For this activity, introduce the directions for students so they are aware of what they are required to do. Release students to complete questions 1 – 3 and review all answers together as a class.



1.6.3 Exploration: Matching Expressions

- Use your calculator to help you match each expression in the first table with an equivalent expression in the second table. Record your answers in the table below.

Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$	$\sqrt[4]{64}$
$(64)^{-\frac{1}{2}}$	$\sqrt[3]{64^2}$
$(64)^{\frac{2}{3}}$	$(\sqrt[3]{64})^4$
$(64)^{\frac{1}{4}}$	$\sqrt{64}$
$(64)^{\frac{1}{2}}$	$(\sqrt{64})^3$
$(64)^{\frac{4}{3}}$	$\frac{1}{\sqrt{64}}$

Rational Exponent	Radical Expression	Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$	$(\sqrt{64})^3$	$(64)^{\frac{1}{4}}$	$\sqrt[4]{64}$
$(64)^{-\frac{1}{2}}$	$\frac{1}{\sqrt{64}}$	$(64)^{\frac{1}{2}}$	$\sqrt{64}$
$(64)^{\frac{2}{3}}$	$\sqrt[3]{64^2}$	$(64)^{\frac{4}{3}}$	$(\sqrt[3]{64})^4$

- Discuss with your partner the relationship between the rational exponent and the radical expression.
Answers will vary. I think it looks like the denominator of the exponent is the same as the index on the radical. I think the negative exponent rule still applies in both situations.
- Work with your partner to complete the following statement.

$$\left(a^{\frac{1}{c}}\right)^b = (a^b)^{\frac{1}{c}} = (\sqrt[c]{a})^b = \sqrt[c]{(a^b)}$$



Teaching one method of relating rational exponents to an equivalent radical expression does not give the student the autonomy of choosing a method that makes sense to them. The flexibility that students gain from knowing that there are multiple ways of representing rational expressions and radical expressions builds fluency.



Question 2 is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.



Spend a little extra time on question 3. In theory, students will have been able to discover this pattern through this activity. Use the time for reviewing the activity and answers to make sure that misconceptions are addressed and their statement for question 3 is correct.

1.6.4 Activity: Card Sort

Component Overview: Students will work with a partner to sort cards with expressions written in rational exponent form and radical expression form into equivalent sets. For this activity, provide students with instructions on what they will be doing and release students to work with their partner to pair them into equivalent pairs.



1.6.4 Activity: Card Sort

1. Work with your partner to sort the cards into sets of equivalent expressions.

$(18^2)^{\frac{1}{3}}$	$(21^{\frac{1}{4}})^5$	$18^{\frac{3}{2}}$	$(21^5)^{\frac{1}{4}}$
$(\frac{1}{21^4})^{\frac{1}{5}}$	$\frac{1}{(\sqrt[4]{21})^{-5}}$	$21^{\frac{5}{4}}$	$(18^3)^{\frac{1}{2}}$
$(18^{\frac{1}{2}})^3$	$(21^{\frac{1}{5}})^{-4}$	$18^{\frac{2}{3}}$	$\frac{1}{(\sqrt[5]{21})^4}$
$21^{-\frac{4}{5}}$	$(\sqrt[3]{18})^2$	$(\sqrt{18})^3$	$(18^{\frac{1}{3}})^2$

Record your matches below.

Equivalent Expression
$(18^2)^{\frac{1}{3}}$ $18^{\frac{2}{3}}$ $(\sqrt[3]{18})^2$ $(18^{\frac{1}{3}})^2$
$(21^{\frac{1}{4}})^5$ $(21^5)^{\frac{1}{4}}$ $21^{\frac{5}{4}}$ $\frac{1}{(\sqrt[4]{21})^{-5}}$
$18^{\frac{3}{2}}$ $(18^3)^{\frac{1}{2}}$ $(18^{\frac{1}{2}})^3$ $(\sqrt{18})^3$
$(\frac{1}{21^4})^{\frac{1}{5}}$ $(21^{\frac{1}{5}})^{-4}$ $\frac{1}{(\sqrt[5]{21})^4}$ $21^{-\frac{4}{5}}$



This activity can be done with the material written in the book by using lists, different markings, or coloring materials to indicate the matching sets. For kinesthetic learners, consider creating actual cards for them to physically compare and match while working on this activity. This can be done by printing the cards and cutting them out or writing each expression onto an index card.



For the whole-group share-out, consider a rapid whiteboard activity. Have a student write one of the expressions and give the rest of the class 30 seconds to write the correct matching expression and hold them up for everyone to see. Discuss the correct answer and repeat this process until all expressions have been completed.



For a challenge, consider having students create expression cards and allowing them time to give those cards to another classmate for them to sort.

1.6.5 Try It: Writing Equivalent Expressions

Component Overview: Students will practice writing equivalent expressions as a rational exponent and radical expression with a partner.



1.6.5 Try It: Writing Equivalent Expressions

1. Complete the table.

$(a)^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
$(100)^{\frac{3}{2}}$	$\left((100)^{\frac{1}{2}}\right)^3$	$((100)^3)^{\frac{1}{2}}$	$(\sqrt{100})^3$	$\sqrt{100^3}$
$60^{\frac{5}{4}}$	$\left((60)^{\frac{1}{4}}\right)^5$	$((60)^5)^{\frac{1}{4}}$	$(\sqrt[4]{60})^5$	$\sqrt[4]{60^5}$
$8^{\frac{3}{5}}$	$\left((8)^{\frac{1}{5}}\right)^3$	$((8)^3)^{\frac{1}{5}}$	$(\sqrt[5]{8})^3$	$\sqrt[5]{8^3}$
$27^{\frac{2}{3}}$	$\left((27)^{\frac{1}{3}}\right)^2$	$((27)^2)^{\frac{1}{3}}$	$(\sqrt[3]{27})^2$	$\sqrt[3]{27^2}$

2. Based on the equivalent expressions in the table, write three equivalent expressions for $(\sqrt{17})^8$.

$$\sqrt{17^8}; 17^{\frac{8}{2}}; \left((17^{\frac{1}{2}})^8\right)$$



Students may need additional support with getting started on the table activity in question 1. Begin by showing students an example or two from the first row. Students may also struggle with how to work backwards in the remaining row. Show students how to work backwards in the second row similarly, so they can better understand how to work through the table.



Student performance on this activity will provide data on which equivalent representation, if any, students struggle with. Consider making notes of where students are struggling to be able to provide them differentiated and targeted support during 1.6.6 Your Turn! or in later lessons. This Try It is helping students extend their fluency of the Laws of Exponents.

1.6.6 Your Turn!

Component Overview: Students will work independently to practice rewriting radical expressions into rational exponent form. For this activity, release students to complete question 1 on their own, and review the answers as a class.



1.6.6 Your Turn!

1. Rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$, where a , b , and c are integers.

$$(\sqrt[3]{72})^5$$

$$\sqrt[5]{5^3}$$

$$\frac{1}{(\sqrt[3]{20})^4}$$

$$\frac{1}{\sqrt[6]{6}}$$



Students who struggled in the prior activity could use this time to get additional support. Consider pulling a small group together to review these questions and use the gradual release model to walk students through the process again in a smaller group. These students may not complete all four of the questions in the allotted time, but their goal will be to understand what they didn't in prior instruction and activities.



For a challenge, consider having students create a problem and trading with another student to have them rewrite it into the other form.

1.6.7 Wrap-Up: Cool Down

Component Overview: Students will complete a cool down that assesses their understanding of equivalent expressions in radical expression in rational exponent form. This activity is designed to be done independently to gather accurate data on student mastery of the content of this lesson. Students should not use a calculator to complete the Wrap-Up.



1.6.7 Wrap-Up: Cool Down

1. Circle the expressions equivalent to $(25^3)^{\frac{1}{2}}$.

$(25)^{\frac{3}{2}}$
 $(\sqrt{25})^3$
 $(25)^{\frac{2}{3}}$
 $(25^{\frac{1}{3}})^2$
 $\sqrt{25^3}$



Student performance on this Wrap-Up can provide data that indicates additional support and differentiation that may be necessary. Use the responses to track any common errors made by the class. Consider using the most common errors in a “my favorite mistake” with the whole class or in small groups.